Neil Christopher M. Rodas

BS Information Technology · Networking, Cloud Security & Edge Computing

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SUMMARY

BS Information Technology student and three-time Dean's Lister who builds systems where networks, security and communities meet, from a solar-powered offline Wi-Fi relay hub to a benchmarked layered cipher. Trained through Cisco CCNA coursework, Google Project Management courses and 13 Google Cloud skill badges.

EDUCATION

Bachelor of Science in Information Technology · Mapúa Malayan Colleges Laguna, Cabuyao, Laguna — 2023 – Present

- **Dean's Lister:** AY 2023–2024, AY 2024–2025 and AY 2025–2026

CAPSTONE AND RESEARCH

A Green Edge Computing-Based Communication Relay Hub via Secured Offline Public Soft Access Point Channel for Community Information Services — AY 2025–2026

- Co-developing a solar-powered, offline point-of-presence node so Barangay Marinig, Cabuyao (64,310 residents) keeps receiving announcements during typhoon power and signal outages.
- Arduino Uno + WeMos ESP32 edge node on a 30W solar panel and 12V 10Ah $LiFePO_4$ battery, broadcasting a SoftAP with Captive Portal, WPA2/WPA3 and AES-128 over about 50–100 m.
- Resident mobile app (distributed by on-site QR code) and browser-based admin web app for announcements, emergency assets and forms.

Performance, Entropy Analysis, and Brute-Force Resilience of a Transposition-Fibonacci Spiral Hybrid Cipher with Caesar Shift Keying — Co-author

- Implemented CipherWeave in Python (Caesar Shift → Columnar Transposition → keyed Fibonacci Spiral) and evaluated keyspace, Shannon entropy and runtime.
- Found a maximum keyspace of about 30 bits, entropy of 3.3869 → 3.4586 bits/char and throughput near 2 MB/s.

ACADEMIC PROJECTS

- **Parkúa: Parking Tracking App and QR Code Scanning for MMCL** (May–Sep 2025, Quality Assurance): QR-based student verification plus real-time, sensor-based parking vacancy; five-person team, ₱102,000 plan.
- **Tenant Information System for S&E Apartments** (Jan 2025): gathered requirements by interview and observation; designed an ASP.NET web system with a normalized relational database.
- **Artelier** (mobile MVP): co-designed an AI-free art platform in Figma for artist portfolios and commissions.
- **Other machine problems:** Poison Game, duoable, Hotel Management System.

TECHNICAL SKILLS

- **Networking & Security:** Cisco switching, routing and wireless; SoftAP and captive portals; GCP VPC and load balancing; AES-128, WPA2/WPA3, access control, cloud security risk management, incident response
- **Development & Hardware:** HTML, CSS, JavaScript, Python, ASP.NET (.NET Framework), relational databases; Arduino Uno, ESP32, solar power systems
- **Tools & Methods:** Google Cloud, Figma, Agile, project planning, quality assurance, SDLC

CERTIFICATIONS

- **CCNA: Switching, Routing, and Wireless Essentials**, Cisco Networking Academy · Nov 2025
- **Google Project Management, 5 courses (Coursera)** · Feb 2026: Foundations, Initiation, Planning, Execution, Agile Project Management
- **13 Google Cloud skill badges (Google Skills)** · Feb–Mar 2026: cloud security risks, threat detection and response, cloud security analyst prep, secure networks, load balancing, app dev environments, ML APIs, and all four Google Cloud Computing Foundations courses

ACTIVITIES AND INTERESTS

- Game Jam participant (2023) · Seminars: Data Security in Computer Programming Languages (EPCPS, Feb 2026); Introduction to Blockchain (JPCS Mapúa MCL, Feb 2025)
- Interests: chess; plays bass, guitar and piano